Advanced Configuration and Power Interface (ACPI) Specification

Release 6.6

UEFI Forum, Inc.

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CONTENTS

List of Tables

List of Figures

Revision History

Many people have contributed to the contents of this specification, including the following:

- ACPI Specification Working Group (ASWG)
- Tianocore Community Members
- Others as noted in the Revision History below

Table 1: **Changes in this release**

Revision #	Issue / Description / Submitter	Modified or Added Content
6.6	M2344 - RAS2 improvements for patrol scrub	Table 5.95, Table 5.99
6.6	M2348 - RISC-V: Add APIC structure in MADT	Table 5.21, Section 5.2.12.27
6.6	M2353 - Adding new registers to the CPPC interface	Section 8.4.6.1, Table 8.23, Section 8.4.6.1.2.3, Section 8.4.6.1.2.6, Section 8.4.6.1.2.7, Table 8.24, Section 8.4.6.1.2.8, Table 8.25, Section 8.4.6.1.2.9, Section 8.4.6.1.3.2, Table 8.26
6.6	M2354 - Describing hot-pluggable memory	Section 9.11, Section 9.11.1, Section 9.11.2, Section 9.11.3, Table 6.48, Section 19.6.45
6.6	M2355 - Reserve "ASPT" signature (AMD Secure Processor Table)	Table 5.6
6.6	M2349 - RISC-V: Add RHCT table	Section 5.2.37, Section 5.2.38, Section 5.2.39
6.6	M2361 - Code first - Exposing Specific Purpose Memory in SRAT	Table 5.69
6.6	M2366 - Code first - ACPI MADT MPWakeup	Section 5.2.12.19, Table 5.46
6.6	M2374 - Mechanism to describe processor power (voltage) planes	Section 6.2.10
6.6	M2375 - Add a new ACPI Firmware Inventory device to the spec	Table 5.244, Section 9.20, Section 9.20.1
6.6	M2379 - Remove IPF support	Section 2.1, Table 5.10, Table 5.13, Table 5.15, Table 5.29, Section 15.4, Section 18.3, Section 5.2.12.11
6.6	M2381 - RISC-V: Add AIA and PLIC APIC structure in MADT	Section 5.2.12, Table 5.21, Table 5.55, Section 5.2.12.28, Section 5.2.12.29, Section 5.2.12.30
6.6	M2382 - RISC-V: Update RHCT table	Table 5.5, Section 5.2.37, Table 5.214, Table 5.215, Section 5.2.38.1, Section 5.2.38.2
6.6	M2385 - SRAT hot-plug memory clarification	Table 5.69
6.6	M2404 - Support for resetting the Multiprocessor Wakeup Mailbox	Section 5.2.12.19, Table 5.45, Table 5.46
6.6	M2405 - Reserve "RQSC" table signature	Table 5.6
6.6	M2406 - Clarify ResourceUsage Descriptor Argument	Section 6.2, Table 6.13
6.6	M2407 - Clarify _CCA on RISC-V	Section 6.2.18
6.6	M2416 - FPDT Add generic Host firmware and microcontroller boot performance records	Section 5.2.24.1, Table 5.123, Section 5.2.24.3, Table 5.124, Section 5.2.24.4, Table 5.125, Section 5.2.24.6, Section 5.2.24.7, Section 5.2.24.8, Table 5.129, Table 5.129, Table 5.130, Section 5.2.24.11, Section 5.2.24.12

continues on next page

Table 1 – continued from previous page

6.6	M2419 - Clarifying the definition of ResourcePriorityRegisters returned via _CPC	Section 8.4.6.1, Table 8.23, Section 8.4.6.1.2.7, Table 8.24, Section 8.4.6.1.2.10
6.6	M2422 - CodeFirst - MADT new GIC flags for non-coherent components	Table 5.19, Table 5.37, Table 5.41, Table 5.42, Table 5.43, Table 5.44
6.6	M2423 - Table name reservation for I/O Resource Director Technology Table (IRDT)	Table 5.6
6.6	M2425 - MADT GICC - Deprecate Parking Protocol for Arm	Table 5.36
6.6	M2429 - Add support PCC Word/DWord/QWord resources, corresponding macros	Table 5.2, Table 6.45, Table 6.46, Table 6.47, Section 19.6.36, Section 19.6.112, Section 19.6.153
6.6	M2430 - Add NHLT table specification	Section 5.2.27
6.6	M2433 - Add RISC-V RINTC Affinity Structure in SRAT	Section 5.2.16, Section 5.2.16.8, Table 5.79, Table 5.80
6.6	M2434 - Typos in ACPI r6.5	Section 9.1.1
6.6	M2450 - New objects for GPE handling in low-power S0 idle	Section 5.6.4.3
6.6	M2458 - _PIC: Add new codes	Section 5.8.1
6.6	M2459 - "Extended-linear" addressing for direct-mapped memory-side caches	Table 5.181
6.6	M2461 - RISC-V: Minor fixes / clarifications on top of M2381 and M2382	Section 5.2.12, Section 5.2.12.27, Table 5.55, Table 5.57, Section 5.2.12.29, Table 5.213, Table 5.214
6.6	M2463 - RAS2 add ADDRESS_TRANSLATION service	Section 5.2.21, Table 5.91, Table 5.98, Section 5.2.21.2.3, Table 5.101, Table 5.102, Table 5.103
6.6	M2472 - Add signature for RISC-V IOMMU Table	Table 5.6
6.6	M2473 - Add FFH reference for RISC-V	Section 8.4.3.2
6.6	M2474 - RISC-V : Clarify IMSIC related fields	Table 5.55, Table 5.57
6.6	M2478 - Add new _PCS and _PST objects to Power Source definition	Table 5.229, Section 10.3.5, Table 10.14, Section 10.3.6
6.6	M2486 - Add signature for LoongArch IOMMU table	Table 5.6
6.6	M2505 - Typos in ACPI r6.6 draft (part 1)	Various locations in the specification.
6.6	M2506 - Typos in ACPI r6.6 draft (part 2)	Various locations in the specification.

Table 2: **Changes in previous releases**

Revision #	Issue # / Description	Modified or Added Content
6.5A	2352 - Clarify use of _DMA without resources	Section 6.2.4
6.5A	2358 - Remove extra text from the definition of PCIEXP_WAKE_DIS in the PM1 Enable Registers	Table 4.12
6.5A	M2383 - Appendix C correction (deprecated content)	Section C
6.5A	2387 - Clarify PCC Type 3 and 4 subspace usage descriptions	Section 14, Section 14.1.6, Section 14.2, Section 14.3, Table 14.13, Section 14.5, Section 14.6, Section 14.6.2
6.5A	2401 - Clarify behavior when _Lxx and _PRW target the same GPE resource	Section 5.6.4.1, Section 5.6.4.2, Table 6.13, Section 7.3.13
6.5A	2402 - ACPI _Sx Support	Section 4.8, Section 4.8.3.7, Section 7.4.2
6.5A	2414 - Clarification of what "Reset End" means	Section 5.2.24.9
6.5A	2420 - Correcting typos in ACPI 6.5	various locations in spec

continues on next page

Table 2 – continued from previous page

6.5A	2432 - Clarify that _IFT and _SRV are used by the DMTF MCTP HI (DSP0256) specification	Table 5.245
6.5A	2435 - EINJv2 Changes	Table 18.25, Table 18.30, Table 18.31, Table 18.32, Section 18.6.4.1, Table 18.34, Table 18.35, Section 18.6.6
6.5A	2442 - Clarifications needed for PDTT section	Table 5.185, Table 5.186, Section 5.2.30.1
6.5A	2444 - Clarify "Global System Interrupt" usage	Throughout spec: replace "GSIV" with "GSI," and capitalize Global System Interrupt(s).
6.5A	2451 - _STA (Device Status) return value info clarification	Section 6.3.7
6.5A	2452 - Remove OpRegion specific text	Section 5.5.2.4.6.1, Section 5.6.8, Section 6.5, Section 6.5.8, Section 9.17.15
6.5A	2469 - Fix description of the OEM Table ID in PPTT	Section 5.2.31
6.5A	2471 - Typos in ACPI Spec 6.5	Various locations in spec.
6.5A	2479 - Fix the order of column 2 and 3 for bitfields of GTDT	Table 5.137, Table 5.141, Table 5.142, Table 5.144
6.5A	2481 - Fix the grammar of the PkgLength PkgLeadByte	Section 20.2.4
6.5	2122 DTPR signature reservation	Table 5.5
6.5	2151 Reserve an _SB._OSC bit and an OperationRegion Subtype for Platform Runtime Mechanism (PRM)	Table 5.221, Table 6.13
6.5	2152 Code first: Add the Virtual I/O Translation (VIOT) Table (Al Stone and others)	Section 5.2.33
6.5	2162 Reserve ACPI table signature for SVKL	Table 5.6
6.5	2177 Reserve ACPI table signature for CCEL	Table 5.6
6.5	2188 Code First: Add 'CXL Root Object' _HID (Vishal Verma)	Section 5.2.6, Table 5.244
6.5	2195 Remove section 9.7 Embedded Controller Device	*Appendix C: Deprecated Content*
6.5	2196 Introduce unaccepted memory type - AddressRangeUnaccepted	Table 15.1
6.5	2198 Clarification to Address Space ID	Table 5.1
6.5	2203 Code First: Add APIC Structures for Loongarch in MADT (LV Jianmin)	Section 5.2.12, Section 5.2.12.20 & sections following.
6.5	2206 Add new PERSISTENT_CPU_CACHES bits to FADT Fixed Feature Flags table	Table 5.10
6.5	2210 Update reference link for the PnP BIOS Spec	Section 6.2.2
6.5	2215 Update to S4 language	Table 5.13, Section 16.1.4.1
6.5	2224 Code First - _DSC Deepest State for Configuration (Rafael Wysocki)	Table 7.3, Section 7.3.27
6.5	2228 Code First - RASF Gen2 (Samer El-Haj-Mahmoud)	Section 5.2, Table 5.5, Section 5.2.21
6.5	2233 Connection Sharing update for Serial Bus Connection Descriptor	Section 6.4.3.8.2.1, Table 6.55, Section 19.6.59
6.5	2236 Code First: Generic Port, performance data for hotplug memory (Dan Williams)	Section 5.2.16, Section 5.2.16.6, Table 5.78, Section 5.2.16.7
6.5	2239 Code First - RAS2 Error Record Local Address to System Physical Address Conversion (Samer El-Haj-Mahmoud)	Table 5.98, Section 5.2.21.2.2

continues on next page

Table 2 – continued from previous page

6.5	2241 Reserve ACPI Device ID for Audio Composition Device	Table 5.244
6.5	2245 Code First - DSD property for uefi-clock-frequency (Samer El-Haj-Mahmoud)	Table 6.39, Section 6.4.3.14, Section 19.6.157
6.5	2248 Adding WDDT name reservation into spec	Table 5.6
6.5	2250 Reserve APMT table name	Table 5.6
6.5	2253 Clarification - Time and Alarm Device methods requirements (Samer El-Haj-Mahmoud)	Section 9.17.2, Section 9.17.5, Section 9.17.6, Section 9.17.7, Section 9.17.8, Section 9.17.9, Section 9.17.10
6.5	2258 Deprecate CDIT/CRAT	*Appendix C: Deprecated Content*
6.5	2261 Reserve KEYP table name	Table 5.5
6.5	2267 Code first - EINJ updates for CXL (Thanunathan Rangarajan)	Table 18.30
6.5	2268 Updated ECR for adding APIC structures for Loongarch in MADT	Section 5.2.12, Section 5.2.12.20 & sections following.
6.5	2272 Code First - Allow FFH OpRegion (Samer El-Haj-Mahmoud)	Table 5.221, Section 5.5.2.4.2, Table 6.13
6.5	2275 MHSP table signature reservation	Table 5.5
6.5	2281 Reserve “AGDI” table signature	Table 5.5
6.5	2285 Code First - MADT GICC new flags (Samer El-Haj-Mahmoud)	Table 5.37
6.5	2287 Code First - EINJv2 (Harb Abdulhamid and others)	Table 18.25, Section 18.6.2
6.5	2293 _ADR and _UPC changes, _PDO addition for USB4 and USB-C	Section 1.10, Table 6.14, Table 9.11, Section 9.12.1, Section 9.12.2
6.5	2294 Reset Reason Health Record	Section 5.2.32.5
6.5	2296 Reserve “NBFT” table signature	Table 5.5
6.5	2297 Miscellaneous GUIDed Table Entries definition	Section 5.2.4, Table 5.5, Section 5.2.34
6.5	2298 Reserve “SWFT” table signature	Table 5.5
6.5	2303 Code First - Armv9 TRBE Support (Thanunathan Rangarajan)	Table 5.36
6.5	2309 Update of FADT Minor Version	Table 5.9
6.5	2312 Update to the HEST table and adding new error source descriptor	Table 18.2
6.5	2314 Code First - Add confidential computing extension for ACPI (Jiewen Yao)	Section 5.2.35, Section 5.2.36
6.5	2316 Add an “attribution” link to the ACPI spec	See top of this Revision History list.
6.5	2322 File name references consistency (upper/lower case)	throughout the spec
6.5	2328 Add ACPI Burst Mode Opt-Out	Table 12.20
6.5	2331 IAPC_BOOT_ARCH’s description in FADT table points to an incorrect table	Table 5.9
6.5	2332 EISAID macro corrections	Section 6.5.11, Section 19.6.65
6.5	2333 USB power data object (_PDO)	Section 9.13
6.5	2334 Power Button Override clarification	Table 4.1
6.5	2335 Comments on review draft	Section 5.2.32.5, Table 5.200
6.5	2338 Table name reservation (IERS)	Table 5.6
6.5	2345 draft spec feedback	Section 5.2.16.6, Table 5.78, and miscellaneous corrections
6.5	2346 Inclusive language update for ACPI spec	Section 1.1.1
6.4 A	2179 _BPT control method: arg2’s description is incomplete in ACPI 6.4 draft	Section 10.2.2.10

continues on next page

Table 2 – continued from previous page

6.4 A	2181 Missing new ACPI 6.4 predefined names in Table 5.173: Predefined ACPI Names	Table 5.245
6.4 A	2187 Some parts of FPDT and SDEV sections should be re-ordered	Section 5.2.28.1
6.4 A	2193 Remove section 9.4	*Appendix C: Deprecated Content*
6.4 A	2194 Remove deprecated content in section 8.4	*Appendix C: Deprecated Content*
6.4 A	2195 Remove section 9.7	*Appendix C: Deprecated Content*
6.4 A	2198 Clarification to Address Space ID	Table 5.1
6.4 A	2211 Two corrections to the Buffer 0 Return Value table	Table 6.4
6.4 A	2216 Incorrect DBPG2 reference	Table 5.6
6.4 A	2219 PPTT is missing in DESCRIPTION_HEADER Signatures for tables defined by ACPI	Table 5.5
6.4 A	2220 Document meaning behind _MEM attributes	Section 6.4.3.5.4.1
6.4 A	2221 Document architecture mapping for extended attributes in Type Specific Attributes	Section 6.4.3.5.4.1
6.4 A	2223 Code First - correct _DMA resource type example	Section 6.2.4
6.4 A	2242 Note misplaced in the Memory Resource Flag Definitions table (resource type=0)	Table 6.49, plus other sections of chapter 6.
6.4 A	2244 shareable (10 used) or sharable (3 used) in ACPI spec 6.4	Table 5.9, Table 5.245, Section 9.12
6.4 A	2254 Incorrect link in 6.4.3.7 Generic Register Descriptor	Section 6.4.3.7
6.4 A	2257 What is meant by handling an error.	Table 18.3
6.4 A	2273 _STA and _DIS Clarifications	Section 6.2.3, Section 6.3.7
6.4 A	2274 Code First - Update HW Error Notification Structure to reference SDEI	Table 18.14
6.4 A	2276 Pin Group Configuration Descriptor: Resource Identifier binary encoding incorrect	Table 6.64
6.4 A	2282 Code first - Fix incorrect reference to "Memory Aggregator Device"	Table 6.64
6.4 A	2283 Code first - BGRT table "valid" field typo	Table 5.120
6.4 A	2284 Inclusive language rename for PCCT sub-space types 3 & 4	Table 14.7, Table 14.12, Table 14.13, Section 14.5, Section 14.6.1, Section 14.6.2
6.4 A	2299 Correction for Device Power Management Objects	Section 7.3
6.4 A	2301 Invalid section reference in CopyObject ASL operator definition	Section 19.6.17
6.4 A	2304 _PLD content missing in table 6.4, spec rev 6.4	Table 6.4
6.4 A	2305 Remove orphaned reference to deprecated PPTT Type 0	Section 5.2.31.1
6.4 A	2307 Missing note numbers in Appendix B table B3	*Appendix B: Video Extensions*
6.4 A	2308 Update/Clarification to _STA	Section 6.3.7
6.4 A	2310 FADT Format clarifications	Table 5.9
6.4 A	2323 Update of FADT Minor Version for ACPI 6.4 Errata A	Table 5.9
6.4	1933 Remove obsolete DDBHandle data type	Section 19, Section 20
6.4	1975 NFIT PMTT Memory Topology	Section 5.2.22.12, Section 9.19.3
6.4	1988 VDIMM SPA Location Cookie	Table 5.147
6.4	1991 Generic Initiator clarifications	Table 5.78, Section 5.2.29.1, and Section 5.2.29.4

continues on next page

Table 2 – continued from previous page

6.4	1997 Add Fuel Gauge Support to Control Method Battery device	Section 10.2, Section 10.2.1, Table 10.3, Table 10.11
6.4	2006 Add _SB._OSC bit for native USB 4 support/control	Section 6.2.12.1.3, Table 6.13, Section 6.2.12.3
6.4	2010 Define new PCC Structure (Type 5)	Section 14.1.7, Section 14.4, Section 5.2.30.1
6.4	2044 Query ARS Capabilities Clarification	Table 9.20
6.4	2045 CXL ACPI enumeration	Table 5.244, Section 6.5.11
6.4	2056 Signature Reservation for Regulatory Graphics Resource Table (RGRT)	Table 5.6
6.4	2070 Define Address encoding for PCI BAR Target GAS structure	Table 5.2, Table 6.13
6.4	2075 Add reference to CDAT Structure from ACPI table	Section 17, https://uefi.org/acpi
6.4	2076 Reserve CEDT signature	Table 5.6, https://uefi.org/acpi
6.4	2077 Clarify CXL _CBR enumeration method	Table 6.71
6.4	2081 Add Connection Descriptor definition and macro for MIPI CSI-2	Table 6.55, Section 6.4.3.8.2.4, Section 19.6.24
6.4	2087 Add MultiProcessor Wakeup structure	Table 5.21, Section 5.2.12.19
6.4	2090 ECR for Battery Charge Limiting (BCL) mode support	Section 3.9.6, Table 6.13, Table 10.10, Table 10.6, Section 10.2.2.5
6.4	2094 New platform telemetry data table - PTDT, reservation and definition	Table 5.5, Section 5.2.32
6.4	2104 Reserve ACPI table signature for the PRMT	Table 5.6
6.4	2105 Increase FADT Major & Minor number to match next ACPI release.	Table 5.9
6.4	2108 Add new ACPI device ID for USB4 host routers	Table 5.244
6.4	2111 Add Access Components for Secure ACPI Enumerated Devices in the SDEV table	Section 5.2.28.1.1
6.4	2118 AEST table signature reservation	Table 5.6
6.4	2120 MPAM Table Name Reservation	Table 5.6
6.4	2121 HMAT updates to support systems with heterogeneous memory	Section 5.2.29.1, Section 5.2.29.4
6.4	2126 Rename SBSA Generic Watchdog and move the spec link to the UEFI website	Section 5.2.25, Table 5.138, and Section 5.2.25.2
6.4	2127 BDAT name reservation	Table 5.6
6.4	2133 Remove reference to DMA Protection Policy Table (DPPT)	Table 5.5
6.4	2137 Extend _DDC to support greater than 256 byte buffer return	*_DDC (Return the EDID for this Device)*
6.4	2138 ACPI-based Identifiers for Caches	Table 5.188, Section 5.2.31.1, Table 5.191, Table 5.192
6.4	2144 Clarify SSDT load order	Section 5.2.11
6.4	2146 Error in the HMAT System Locality Latency and Bandwidth Information Structure	Table 5.180
6.4	2150 Clarify description of CoordType in _PSD object	Table 8.1, Table 8.3, Table 8.19, Table 8.22
6.4	2156 Corrections to the FPDT	Fig. 5.8, Table 5.121, Section 5.2.24.1, Section 5.2.24.2, Section 5.2.24.3, Section 5.2.24.4, Section 5.2.24.5, Section 5.2.24.8, Section 5.2.24.9, Section 5.2.24.10
6.4	2157 Processor object cleanup missed ProcessorObj in ObjectTypeKeyword list	*ObjectTypeKeyword*, Table 19.36

continues on next page

Table 2 – continued from previous page

6.4	2159 6.3A contains incorrect heading levels for some sections	various sections
6.4	2162 Reserve ACPI table signature for the SVKL	Table 5.6
6.4	2169 IRQ macro description incorrectly refers to the IO macro	Section 19.6.67
6.4	2170 Feedback on the 6.4 draft	multiple sections; see Mantis for details
6.4	2171 Heading changes for consistency in section 19.6	Section 19.6.103, Section 19.6.104, Section 19.6.105, Section 19.6.106, and Section 19.6.107
6.4	2176 EINJ: Correction for GET_COMMAND_STATUS Action.	Table 18.18
6.4	2179 _BPT control method: arg2 description is incomplete in ACPI 6.4 draft	Section 10.2.2.10
6.4	2180 New section from ECR M2010 misplaced in ACPI 6.4 draft	Section 14.1.7
6.4	2181 Missing new items in ACPI Predefined Names table	Table 5.245
6.4	2182 Multiprocessor Wakeup Structure misplaced in spec	Table 5.21, Section 5.2.12.19
6.4	2183 Incorrect PHAT reference in Table 5-5	Table 5.5
6.4	2186 Error in sample code	Section 5.6.9.2, Section 5.6.9.3
6.4	2187 Some parts of SDEV sections should be reordered	Section 5.2.28.1
6.4	2191 Feedback on ACPI 6.4 draft	*various sections*
6.4	2197 Typos in the t-state dependency and p-state dependency tables	Table 8.19, Table 8.22
6.3 A	1952 Serious issues with Generic Serial Bus chapters	Section 5.5
6.3 A	1972 Add links to grammar definitions	Section 19.2, Section 20.2, Section 21.2.2
6.3 A	1973 Change name of TypeXOpcodes for clarity	Section 19.2, Section 20.2
6.3 A	1977 Errata for GHES_ASSIST (APEI) feature	Table 18.3, Table 18.5, Table 18.10, Table 18.15, and Section 18.7
6.3 A	1981 Minor issues with BGRT description and field names.	Table 5.120
6.3 A	1985 ASL macro definitions reversed between "For" and "Fprintf"	Section 19.3.4
6.3 A	1990 _PR0 fixes	Section 7.3.8
6.3 A	1995 Clarifcation to the Guaranteed Performance Register implementation	Section 8.4.6.1.1.6
6.3 A	2001 Clarifications for PCI Express AER ownership	Section 18.3.2.4, Section 18.3.2.5, Section 18.3.2.6
6.3 A	2004 Appendices numbering	*Appendix A: Device Class Specifications*, *Appendix B: Video Extensions*, *Appendix C: Deprecated Content*
6.3 A	2011 _DSD link in Generic Buttons Device Child Objects table	Section 9.18
6.3 A	2012 Clarify allowed values for ACPI0007 _UIDs	Section 5.2.12, Section 6.1.12
6.3 A	2021 Typo in PM_TMR_BLK field	Table 5.9
6.3 A	2022 Errors in description of "X_GPE0_BLK"	Table 5.9
6.3 A	2037 Incorrect reference in Real Time Clock Alarm	Section 4.8.2.4
6.3 A	2047 Clarifications and Fixes to the Error Injection (EINJ) section	Section 18.6

continues on next page

Table 2 – continued from previous page

6.3 A	2052 Clarify behavior of PerformanceLimitedRegister in _CPC	Section 8.4.6.1.3.2
6.3 A	2057 Clarify wording of delivered performance constraints in CPPC	Section 8.4.6.1.3.1
6.3 A	2059 EISAID Macro - missing algorithm	Section 19.3.4
6.3 A	2064 Make "DPA" definition more generic	*Device Physical Address (DPA)*, Section 9.19.7.8, Section 9.19.7.8.3
6.3 A	2067 Clarify _HID and _ADR usage	Section 6, Section 6.1, Section 6.1.1, Section 6.1.2, Section 6.1.5
6.3 A	2069 Update figure OSPM/ACPI Global System	Fig. 1.1
6.3 A	2072 Deprecate "PPTT Type 2 - Processor ID" section	Was section 5.2.29.3 in ACPI Spec 6.3
6.3 A	2098 Clarification of supported ACPI platform implementations	Table 3.3
6.3 A	2100 Correction/Clarification of _CBA description	Table 5.245
6.3 A	2109 Incorrect SLIT reference in "DESCRIPTION_HEADER Signatures for tables defined by ACPI"	Table 5.5
6.3 A	2112 _TZP questions and issues	Section 11.4.26
6.3 A	2113 Label tables in the OS Initiated section of Idle State Coordination	Section 8.4.3.2.2, Section 8.4.3.2.2.1
6.3 A	2115 Duplicate definition of RawDataBufferTerm	Section 19.2.6
6.3 A	2123 Interrupt Polarity _LL values do not agree between chapters	Section 19.6.65 and Section 19.6.67
6.3 A	2128 Some changes from ECR 1588 are missing in ACPI 6.3	Section 19.6.65
6.3 A	2140 Incorrect offsets in PCC Subspace Structures type 3 and 4	Table 14.7
6.3 A	2141 Typos in Chapters 5 and 17	Revision History, Table 5.23, Section 17.3.1, and Section 17.4.1
6.3 A	2145 Error in the PCC Type 3 and 4 subspace description	Table 14.7
6.3	1851 Extend GTDT to describe ARMv8.1 architected CNTHV timer	Section 5.2.24
6.3	1855 ARS Error Inject	Table 9-299, Section 9.20.7.7, Section 9.20.7.9.1, Section 9.20.7.12
6.3	1867 Add Trigger order to PCC Identifier structure within PDTT	Section 5.2.28
6.3	1873 Peripheral-attached Memory	Table 5-132
6.3	1883 Reserve the table names "CRAT" and "CDIT"	http://uefi.org/acpi
6.3	1893 New NVDIMM Device methods _NCH and _NBS	Section 9.20.8.1, Section 9.20.8.2
6.3	1898 PCC Operation Region	Section 5.5.2.4, Section 6.5.4, Section 19.2.7, Section 19.6, Section 20.2.5.2
6.3	1900 I3C host controller support	Table 6-190, Table 6-241
6.3	1904 Generic Initiator Affinity Structure	Section 5.2.16
6.3	1910 NVDIMM Address Range Scrubbing (ARS) interface update	Section 5.6.6, Section 9.20.7
6.3	1911 _PRD object in Table 6-186 has no definition	Appendix C
6.3	1913 New NVDIMM Device methods for Health Error Injection	Section 5.6.6, Section 9.20.8
6.3	1914 HMAT Enhancements	Section 5.2.27
6.3	1922 _HPX Enhancements	Section 6.2.9

continues on next page

Table 2 – continued from previous page

6.3	1930 ASL: Make some arguments to ASL operators optional	Section 19.6.7, Section 19.6.46, Section 19.6.63, Section 19.6.88
6.3	1931 ASL: extend Load() operator to allow table load from an ASL buffer	Section 19.6.76
6.3	1932 ASL: deprecate Unload operator	Section 19.6.146 and related references
6.3	1934 SPE support for ARM	Section 5.2.12.14, Table 5-155
6.3	1939 Error Disconnect Recover Notification	Table 5-165, Section 6.3.5
6.3	1944 Outdated copied text from PCI Firmware Spec	Section 6.2.11.3, Section 6.2.11.4
6.3	1946 Generic Initiator _OSC Bit	Section 5.2.16.6, Table 6-200
6.3	1948 Adds an “Online Capable” flag to the Local APIC, Local SPAPIC, and x2APIC structures in MADT	Tables 5-46, 5-47, 5-55, and 5-58
6.3	1958 PCC Operation Region Updates	Section 5.5.2.4, Section 19.2.7, Table 19-420, Section 20.2.5.2
6.3	1959 Update to ECR 1914	Table 5-146
6.3	1978 GT Block Timers table - update the Timer Interrupt Mode description	Table 5-126
6.3	1979 ACPI version change from 6.2 to 6.3	Table 5-33
6.3	1980 Fix link to local APIC flags in the Processor Local APIC Structure table	Table 5-46
6.2 B	1819 Errata: remove support for multiple GICD structures	Table 5-43
6.2 B	1852 Fix Inconsistent TranslateType Language	Section 19.6.33, Section 19.6.34, Section 19.6.41, Section 19.6.42, Section 19.6.109, Section 19.6.110, Section 19.6.151
6.2 B	1870 PPTT Clarifications	Section 5.2.29.1
6.2 B	1881 Incorrect reference “Memory Devices” in “5.2.21.10 Interaction with Memory Hot Plug”	Section 5.2.21.10
6.2 B	1882 Incorrect EINJ table references/link	Table 18-404
6.2 B	1894 SRAT GICC Flags Field Definition Errata	Table 5-76
6.2 B	1905 Missing description in 6.1.9 title in ACPI 6.2a	Section 6.1.9
6.2 B	1909 Update NFIT SPA Range Structure	Table 5-132
6.2 B	1929 Miscellaneous Errata	Section 19.6.38, Section 19.6.53, Section 19.6.54, Removed redundant Interrupt section (now Section 19.6.63)
6.2 B	1945 NFIT_SPA_ECR	Section 5.2.25.2
6.2 B	1951 _PXM Clarifications	Section 5.2.16, Section 5.2.16.6, Section 6.2.14, Section 6.2.15, Section 17.2, Section 17.2.1, Section 17.3, Section 17.3.1, Section 17.4, Section 17.4.1
6.2 B	1960 PWR_BUTTON description should say “power button”, not “sleep button”	Table 5-34
6.2 B	1962 Clarifications for the use of _REG methods	Section 6.5.4
6.2 B	1965 Clean up Address Space ID	Table 5-25, Table 6-238, Section 19.6.114, Section 19.2.7
6.2 B	1968 Clarifications for ACPI NamePaths	Section 5.2
6.2 A	1839 Missing space in title of ACPI RAS Feature Table (RASF)	Section 5.2, Section 5.2.20, Table 5-29
6.2 A	1837 Typos in Extended PCC subspaces (types 3 and 4)	Section 14.1.6
6.2 A	1831 Add a new NFIT Platform Capabilities Structure	Section 5.2.25.1, Figure 5-22, Table 5-131, Section 5.2.25.9

continues on next page

Table 2 – continued from previous page

6.2 A	1827 PPTT ID Type Structure offsets	Section 5.2.29.3
6.2 A	1825 Remove bits 2-4 in the Platform RAS Capabilities Bitmap table	Section 5.2.20.4
6.2 A	1820 Region Format Interface Code description	Section 5.2.25.6
6.2 A	1819 Remove support for multiple GICD structures	Section 5.2.12, Section 5.2.12.1
6.2 A	1814 PDTT typos and PPTT reference	Revision History, Section 5.2, Section 5.2.28
6.2 A	1812 Minor correction to Trigger Action Table	Section 18.6.4
6.2 A	1811 General Purpose Event Handling flow	Section 5.6.4
6.2	1795 ACPI Table Signature Reservation	Table 5-30
6.2	1781 Clarify ResourceUsage Descriptor Argument	Table 6-193
6.2	1780 Add DescriptorName to PinFunction and PinConfig Macros	Section 19.6.102 and Section 19.6.103
6.2	1770 Update Revision History	Revision History
6.2	1769 FADT Format: ACPI Version update to reflect 6.2 versus 6.1	Table 5-33
6.2	1755 Deprecate PCC Platform Async Notifications	Section 14.4, and Section 14.5.1
6.2	1743 PinGroupFunctionConfig resource descriptors update	Section 6.4.3.11, Section 6.4.3.12, Section 6.4.3.13
6.2	1738 PCIEXP_WAKE Bits description updates	Table 4-15, Table 4-16, and Table 5-34
6.2	1731 Software Delegated Exception HW error notitication	Section 18-394
6.2	1725 NVST Updates - NFIT ARS Error Injection	Section 9.20.7.9, Section 9.20.7.10, and Section 9.20.7.11
6.2	1724 NVST Updates - Platform RAS Capabilities Updates	Section 5.2.20.4
6.2	1723 NVST Updates - Translate SPA DSM Interface	Section 2.1, Section 9.20.7.8
6.2	1722 NVST Updates - ARS Updates	Section 2.1, Section 9.20.7.2, Section 9.20.7.4, Section 9.20.7.5, and Section 9.20.7.6
6.2	1721 NVST Updates - Labels	Section 2.1, Section 5-184, and Section 6.5.10
6.2	1717 ASL Grammar Update for Reference Operators	Section 19.2
6.2	1714 Reserve the table name “SDEI”	Table 5-30
6.2	1705 Add Heterogeneous Memory Attributes Tables (HMAT)	Section 5.2, Section 5.6.6, Section 5.6.8, Section 6.2, Section 6.2.18, and Section 17.4
6.2	1703 Time & Alarm Device _GCP new bits	Section 9.18.2
6.2	1680 Pin Group, Pin Group Function and Pin Group Configuration Descriptors and Macros	Table 6-224 and Section 6.4.3.10
6.2	1679 Pin Configuration Descriptor and Macro	Table 6-224 and Section 6.4.3.10
6.2	1677 CPPC Registers in System Memory	Section 6.2.11.2 and Section 8.4.7.1
6.2	1674 GHES_ASSIST Proposal	Section 18.3.2
6.2	1669 FADT HEADLESS flag should be valid for HW_REDUCED_ACPI platforms	Section 5.2.9
6.2	1667 Processor Properties Topology Table (PPTT)	Section 5.2.29
6.2	1659 Master Slave PCC channels	Chapter 14, Platform Communications Channel (PCC)
6.2	1656 SRAT Support for ITS	Section 5.2.16
6.2	1650 CPPC Support for Multiple PCC Channels	Table 6-200 and Section 8.4.7.1.9
6.2	1649 ECR: Minor updates to IA-32 Architecture Deferred Machine Check	Section 18.3.2.10
6.2	1645 Add _STR Support for Thermal Zones	Section 6.1, Section 6.1.10, Section 11.4, Section 11.4.14, and Section 11.7.1

continues on next page

Table 2 – continued from previous page

6.2	1632 Secure Devices Table (SDEV)	Table 5-30
6.2	1611 Add a _PPL object to processor devices	Section 8.4.7
6.2	1597 ASL For() Conditional Loop Macro	Section 19.6.51, Section 19.2.5, Section 19.2.6, and Section 19.3.4
6.2	1588 Clarification on Interrupt Descriptor Usage for “Interrupt Combining”	Section 6.2.11.2, Section 6.4.3.6, Section 19.6.62
6.2	1585 Reserve table signature “WSMT, ” with reference to ACPI links page for more details	Table 5-30
6.2	1583 Diverse Highest Processor Performance	Table 5-158 and Table 6-200
6.2	1578 Function Config Descriptor and Macro	Table 6-213 and Section 6.4.3.9
6.2	1576 Platform Debug Trigger Table (PDTT)	Section 5.2.28
6.2	1573 Extensions to the ASL Concatenate operator	Section 19.2.6 and Section 19.6.12
6.2	1569 Add new introduction (background) section	Background chapter
6.1 Errata A	1796 Clarify that Type 1 can never support Level triggered platform interrupt	Section 14.1.4
6.1 Errata A	1785 Lack of clarity on use of System Vector Base on GICD structures	Section 5.2.12.15
6.1 Errata A	1783 Clarification on Interrupt Descriptor Usage for Bit [0] Consumer/Producer	Table 6-237
6.1 Errata A	1760 Typo - incorrect bit offsets in the PM1 Enable Registers Fixed Hardware Feature Enable Bits table.	Table 4-16
6.1 Errata A	1758 Minor Errata in ERST tables, Serialization Instruction Entry and Injection Instruction Entry.	Table 18-399 and Table 18-405
6.1 Errata A	1756 Errata: Ensure non-secure timers are accessible to non-secure in the Flag Definitions: Common Flags table.	Table 5-126
6.1 Errata A	1740 Errata in section 9.13: wrong reference	Section 9.13
6.1 Errata A	1715 0 is a valid GSIV for the secure EL1 physical timer in GTDT	Table 5-120
6.1 Errata A	1687 Typo in the Reserved field of the GIC ITS Structure table.	Table 5-66
6.1 Errata A	1686 Clarification of the FADT HW_REDUCED_ACPI flag description in the FADT Format table.	Table 5-33
6.1 Errata A	1676 Clarifications for the ASL Buffer (Declare Buffer Object)	Section 19.6.10
6.1 Errata A	1671 Typo in Memory Affinity Structure table	Section 5-72
6.1 Errata A	1670 Update for _OSI return value	Section 5.7.2
6.1 Errata A	1664 Clarification of the RSDP Structure table, Revision description.	Table 5-66
6.1 Errata A	1662 Clarification of the Generic Communications Channel Command Field table.	Table 14-370
6.1 Errata A	1661 typos in the Generic Communications Channel Status Field table and the Platform Notification section.	Table 14-371 and Section 14.5
6.1 Errata A	1660 type in the Generic Communications Channel Shared Memory Region table	Table 14-369
6.1 Errata A	1651 LPI Clarifications	Section 8.4.4.3

continues on next page

Table 2 – continued from previous page

6.1 A	Errata	1644 Mismatch of mantis number 1449 vs. change description	Revision History
6.1 A	Errata	1643 Incorrect row order in GET_EXECUTE_OPERATION_TIMINGS table	Table 18-397
6.1 A	Errata	1642 Clarifications and fixes to _PSD and _TSD	Table 5-184
6.1 A	Errata	1639 _WPC and _WPP are missing in the Predefined ACPI Names table.	Table 5-164
6.1 A	Errata	1616 Clarify which processor ID to use in the EINJ for ARM	Table 18-403
6.1 A	Errata	1606 Errata: typos in the Interrupt Resource Descriptor Macro definition	Section 19.6.62
6.1 A	Errata	1602 Updates to the PMC Method Result Codes table	Table 10-338
6.1 A	Errata	1601 Typos in the _CPC Implementation Example	Section 8.4.7.1.11
6.1 A	Errata	1600 Typos in PCC Subspace Structure Type 1 and Type 2.	Table 14-366 and Table 14-367
6.1 A	Errata	1599 Add clarification to existing text (_OSC Control Field via arg3)	Table 6-202
6.1 A	Errata	1591 ASL grammar clarification for "executable" AML opcodes	Section 5.4
6.1 A	Errata	1589 Wireless Power Calibration Device ACPI ID not defined	Section 10.5 (Table 10-292 removed) and Table 5-163
6.1 A	Errata	1582 Clarification for Time and Alarm wake description	Section 9.18.1
6.1 A	Errata	1581 Processing Sequence for Graceful Shutdown Request - need to update section 6.3.5.1 to reflect change	Table 5-166 and Section 6.3.5.1
6.1 A	Errata	1579 typos	Table 5-130 and Table 5-131
6.1 A	Errata	1577 BGRT Image Orientation Offset	Table 5-107
6.1 A	Errata	1572 Update ASL grammar to support multiple Definition Blocks	Section 19.2.3
6.1 A	Errata	1571 Update AML Filename description for ASL DefinitionBlock operator	Section 19.6.28
6.1 A	Errata	1552 GIC Redistributor base address language in GICC leaves room for ambiguity	Table 5-60
6.1 A	Errata	1549 Errata: wrong offset in Generic Communications Channel Shared Memory Region table.	Table 14-369
6.1		1527 Qualcomm feedback on ACPI 6.1 draft 2	Throughout
6.1		1524 Strange hotlink	Section 5.7.5
6.1		1514 Comments against 6.1 Draft from HPE	Minor corrections and fixed typos throughout document, especially Section 9.20.7.2
6.1		1512 Microsoft feedbacks on ACPI 6.1 draft 2	Section 5.2.25, Section 9.20.7, Section 18.3.2
6.1		1503 Editorial comments against 6.1 Draft 1	Throughout–draft corrections & typos
6.1		1500 ACPI 6.1 - Graceful Shutdown (Device Object Notification)	Table 5-166
6.1		1499 _FIT and _MAT ASL nits in 6.0 and 6.1 Draft	Section 6.2.10, Section 6.5.9
6.1		1490 ACPI Version update to reflect 6.1 versus 6.0	Table 5-33

continues on next page

Table 2 – continued from previous page

6.1	1483 NFIT SPD extensions and clarifications	Section 5.2.25x, Section 6.5.9, Section 9.20x
6.1	1478 Wireless Power Calibration ACPI Device	Section 10.5 & Section 10.6
6.1	1427 Addition to Memory Device State Flags in NFIT	Table 5-133
6.1	1395 _DSM interfaces associated with NVDIMM-N objects	Section 9.20.2x through Section 9.20.7
6.1	1384 ERST/EINJ max wait time	Table 18-397, Table 18-404
6.1	1367 Interrupt-signaled Events	Section 4.1.1.1 Section 5.6, , Section 5.6.10, Section 5.6.4, Section 5.6.5 Section 5.6.5.2, Section 6.2.11.2, Section 7.3.13, Section 18.3.2.7.2, Section 18.4, and added
6.1	1356 ARM APEI extensions	Section 18.3.2.7, Section 18.3.2.8, Section 18.3.2.9
6.1	1326	Section 2.2, Table 5-37, Section 7.4.2.5, Section 15, Table 15-374, Section 16.1.4
6.0 Errata	1488 Typo on description of PkgLength encoding (ACPI v6.0, section 5.4)	Section 5.4
6.0 Errata	1487 The Length of GIC ITS Structure is wrong	Table 5-66
6.0 Errata	1470 Region Format Interface Code clarification	Table 5-137
6.0 Errata	1462 5.2.21 Errata	Section 5.2.21
6.0 Errata	1461 5.2.21.10 Clarification	Section 5.2.21.10
6.0 Errata	1449 Graceful Shutdown Request (Device Object Notification Values)	Section 2.1, Table 5-44, Section 5.2.12.6, Table 5-51, Section 5.2.12.9, Section 5.2.12.14 through Section 5.2.12.18, Section 5.2.25, Section 5.6, Table 6-193, Table 6.2.10, Table 6-249, Table 6.5.9
6.0 Errata	1445 Section 19.6.99 “Package” of the specification needs updating	Section 19.6.100
6.0 Errata	1444 GTDT CntReadBase Physical address should be optional	Section 5.2.24
6.0 Errata	1433 Time and Alarm _GCP changes in support of wakes from S4/S5	Section 9.18.2
6.0 Errata	1432 Errata - Explicit Data Type Conversions	Section 19.3.4, Section 19.3.5.2, Section 19.3.5.3
6.0 Errata	1406 NFIT RAMDisk Update	Section 5.2.25.2
6.0 Errata	1403 Two distinct definitions of the MADT have the same revision number	Table 5-43
6.0 Errata	1393 In FADT: if X_DSDT field is non-zero, DSDT field should be ignored or deprecated	Table 5-33
6.0 Errata	1392 Incorrect length in the GIC ITS Structure	Table 5-66
6.0 Errata	1386 Clarify APEI vs UEFI runtime variable support	Table 18-397
6.0 Errata	1385 ACPI 6.0 typo and table misnumbering	Section 18.5.2.1
6.0 Errata	1380 Unnecessary restrictions to FW vendors in ordering of GIC structures in MADT	Section 5.2.12.14
6.0 Errata	1378 Duplication of table 5-155/156, section mismatch in GIC redistributor	Table 5-175 & Table 5-180 duplicates removed, Section 5.2.12.17
6.0 Errata	1374 section mismatch: _CCA method belongs to section 6.2 Device Configuration Objects?	Table 6-189/Table 6-193
6.0 Errata	1372 Fix inconsistency for _PXM method in section 17	Section 17.2.1, Section 17.3.2

continues on next page

Table 2 – continued from previous page

6.0 Errata	1368 Various errata fixes and clarifications in chapter 18 APEI	Section 18.3.1, Section 18.3.2.7.1, Section 18.5.1, Section 18.6.1, Section 18.6.2, Section 18.6.4
6.0 Errata	1361 Clarify _PIC Method on ARM	Section 5.8.1
6.0 Errata	1289 replace use of the term "BIOS" with more accurate descriptions	Throughout spec
6.0 Errata	1154 Ensure that ACPI and UEFI specs agree on the treatment of "holes" in the memory map	Section 15.4
6.0	1344 Sharing of Connection Resources, NOTE: The changes were included in ACPI 6.0, but was missed in the ACPI 6.0 Revision History	Section 5.5.2.4.6 through Section 5.5.2.4.6.3.9 Section 19.6.15
6.0	1370 Changes needed for ACPI 6.0: persistent memory S4 behavior	Section 16.3.4
6.0	1359 Vendor Range for E820 Address Types and UEFI memory Types	Table 15-374
6.0	1354 Disambiguation of _REV	Section 5.7.4
6.0	1343 Comments against v6.0 Final Draft	Section 18.6.2, Section 18.6.4
6.0	1340 comment against the Final Draft: Minor errata in register fields of LPI example	Section 8.4.4.3.4
6.0	1332 Fixes for ACPI 6.0 Draft March 2	Table 5-37, Section 5.2.25.2, Table 5-132
6.0	1328 ACPI 6.0 Draft feedback - Mantis 1228	Table 5-62
6.0	1337 Missing reference in Extended Address Space Descriptor Definition, Section 6.4.3.5.4	Section 6.4.3.5.4
6.0	1333 ACPI 6.0 March2 Draft Feedback - Bits and NFIT related	NFIT throughout
6.0	1329 ACPI 6.0 Feb 18 Draft - Follow consistent notation for Bits and Bytes ranges	throughout
6.0	1327 ACPI 6.0 Feb 18 draft feedback - NFIT related	NFIT throughout
6.0	1324 ACPI 6.0 Feb 5 Draft1 Feeback2 - Mantis 1250	Section 5.2, Section 5.2.25, Section 6.1.1, Section 5.6.6
6.0	1320 ACPI 6.0 Feb 5 Draft1 Feedback - Mantis 1250	Section 5.2, Section 5.2.25, Section 6.1.1, Section 5.6.6
6.0	1319 Comment against ACPI 6.0 Draft 1 concerning Mantis 1279	Section 19.1, Section 19.6.3, Section 19.6.5, Section 19.6.26, Section 19.6.31, Section 19.6.60, Section 19.6.61Section 19.6.68 - Section 19.6.74, Section 19.6.78Section 19.6.85, Section 19.6.86, Section 19.6.92
6.0	1312 Add USB-C Connection support to _UPC	Table 9-293, Section 9.14
6.0	1306 New ACPI Version Placeholder	Table 5-33
6.0	1302 Errata on reference in section 6.2.11.2 Platform-Wide OSPM Capabilities	Section 6.2.11.2
6.0	1294 Typo in section 5.7.2: "Section" used when "Table" was meant	Section 5.7.2
6.0	1293 Reserve "STAO" and "XENV" table signatures	Table 5-30
6.0	1292 A Missing space in first paragraph of Section 2.4	Section 2.4
6.0	1284 Battery ACPI ECR	Section 5-184, Section 10.2.2.7, Table 10-329, Section 10.2.2, Table 10-331
6.0	1282 AML: Improve Disassembly of Control Method Invocations	Section 19.6.44, Section 20.2.5.2, Section 20-440
6.0	1281 ASL Printf and Fprintf Debug MacrosTable 10-331Table 10-331	Section 19.2.5, Section 19.2.6, Section 19.3.4, Section 19.3.5.2, Section 19.3, Section 19.4, Section 19.6.52, Section 19.6.107

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Table 2 – continued from previous page

6.0	1280 ASL Helper Macro for _PLD (Physical Location of Device) - ToPLD()	Section 19.2.6, Section 19.3.4, Section 19.3.5.2, Section 19.4, Section 19.5, Section 19.6.140
6.0	1279 ASL Extensions for Symbolic Operators and Expressions (ASL 2.0)	Section 19.1, Section 19.6.3, Section 19.6.5, Section 19.6.26, Section 19.6.31, Section 19.6.60, Section 19.6.61, Section 19.6.68 - Section 19.6.74, Section 19.6.78, Section 19.6.85, Section 19.6.86, Section 19.6.92
6.0	1265 Missing word in figure 1-1	Figure 1-1
6.0	1264 Device Power Management Clarifications	Section 2.3, Section 2.3.1, Section 3.3.1, Section 3.3, Section 3.4, Section 3.4.2, Section 3.4.3, Section 3.4.3, Section 3.4.4x), Section 7, Section 7.1, Section 7.2x, Section 7.3
6.0	1262 New Thermal Zone Objects	Table 5-184, Section 11.1.5.1, Section 11.4.8, Section 11.4.21
6.0	1261 _OSC, add OS–>Platform information to communicate >16 p-states are supported	Table 6-200
6.0	1258 Standby Thermal Trip	Section 11.4.5
6.0	1253 Clarification of S5 (Soft-Off) and S1~S4 Sleeping States	Section 2.4, Section 3.9.4, Section 4.7, Section 4.8.2.3, Section 4.8.3.2.1, Section 7.3.1
6.0	1252 Incorrect Indentation in first page of Section 3	Section 3
6.0	1250 Support for Non-Volatile Memory Firmware Interfaces	Section 5.2, Section 5.2.25, Section 6.1.1, Section 5.6.6
6.0	1241 PCC and level interrupts for HW reduced platforms	Section 14.1.2, Section 14.1.5
6.0	1232 Deprecate Processor Keyword	Table 5-46, Table 5-52, Section 5.2.12.10, Section 5.2.12.12, Section 8.4, Section 11.7.1, Section 11.7.2, Section 19.6.30, Section 19.6.108
6.0	1231 Adjust max p-states	Section 2.6
6.0	1230 Adding Support for Faster Thermal Sampling	Table 6-200, Table 5-184, Section 11.4.17, Section 11.4.22, Section 11.6
6.0	1229 Reserve IORT and support for ARM GICv3/4 ITS in MADT	Table 5-29, Table 5-45, Section 5.2.12.18
6.0	1206 Clarify _HID/_CID/_CLS usage model	Section 6.1, Section 6.1.5, Section 6.2x
6.0	1203 CPPC heterogeneous performance capabilities	Section 8.4.7, Section 8.4.7.1.10
6.0	1197: MADT Efficiency Classes and wording change for MP Parking update	Table 5-60
6.0	1176 FADT Hypervisor Vendor Identification Support	Table 5-33
6.0	1171 Extend Address Ranger Types and UEFI Memory Type to comprehend persistent memory.	Table 5-37, Section 6.4.3.5.4.1, Section 15, Table 15-379, Section 15.4, Table 15-380
6.0	1152 Support for Platform-specific device reset	Section 7.3.25 and Section 7.3.26 t, Table 7-255 Table 7-256
6.0	1132 Generic Button(s) Abstraction	Table 5-183, Add new Section 9.19 and following
6.0	1125 ACPI Low Power Idle Table (LPIT) and _LPD proposal	Section 5.6.7, Section 5.6.8, Table 6-200, Section 7.1, Section 7.2.5, Section 7.4.2.1, Section 8.4, Section 8.4.1, Section 8.4.2, Section 8.4.2.1, Section 8.4.3.1
5.1 Errata	1265 Missing word in figure 1-1	Figure 1-1
5.1 Errata	1252 Incorrect Indentation in first page of Section 3	Section 3

continues on next page

Table 2 – continued from previous page

5.1 Errata	1243 Clarify whether or not the FACS is optional or not	Section 5.2.9, Table 5-33
5.1 Errata	1233 Fix broken Link and Example for _CLS	Section 6.1.3
5.1 Errata	1228 Present GIC version in MADT table	Table 5-62
5.1 Errata	1196 Table reference in Section 9.8.3.2 is Incorrect	Section 9.9.3.2
5.1 Errata	1193 Parking protocol field link is incorrect	Section 5.2.12.14, Table 5-60
5.1 Errata	1190 Table references in Section 18 - ACPI Platform Error Interfaces (APEI) are incorrect	Table 18-383, Table 18-385
5.1 Errata	1189 _CCA attribute default value description does not work for ARM systems	Section 6.2.17
5.1	1181 MADT GICC table definition is wrong	Table 5-61, 5.2.12.14
5.1	1180 FADT minor version byte length is wrong	May-34
5.1	1179 Errors in GTDT Section of 5.1 draft	5.2.24, 5.2.24.1, Tables 5-115, 5-118, 5-121, 5-122
5.1	1175 Bad section reference in ACPI 5.1	19.2.3
5.1	1164 Modifications to UEFI Forum ownership of PNP ID and ACPI ID Registry	6.1.5
5.1	1161 Misc typos in draft documents	5.2.1.6, 5.2.16.4, 5.2.24, 5.2.12.14, 5.2.24.1.1, Table 5-74, Table 5-115-116, Table 5-118-119, Table 5-121, Table 5-61, 5-61 8.4.5.1, 8.4.5.1.2.3 Table 6-162, Table 8-229, RM duplicates from 1123/1130:8.4.5.1.31.1
5.1	1160 ACPI 5.1 draft corrections related to _DSD (SEE #1126)	6.2.5, Table 5-148 & 6-157
5.1	1157 Reserve ACPI Low Power Idle Table Signature “LPIT”	Table 5-31
5.1	1155 Updates to M1133 MADT	Table 5-63, 5-64
5.1	1151 Bug in ASL example code	PRT3 code example following Figure 9-49
5.1	1149 GTDT changes for new GT Configurations	5.2.24, 5.24.1x
5.1	1136 Add a Notification Type for System Resource Affinity Change Event	Table 5-119 Device Object Notifications, new 17.2.2
5.1	1134 FADT changes for PSCI Support on ARM platforms	Table 5-34, 5-36, New table 5-37
5.1	1135 PCC Doorbell Protocol for HW-Reduced Platforms	14.1.1, 14.1.2-4, 14.2.1-2, 14.3-4
5.1	1133 MADT Updates for new GICs	5.2.12.15-17, Table 5-43, 5.2.12 table 5-45, 5-60, 5-61, 5-63, 5-66
5.1	1131 Per-device Cache-coherency Attribute	6.2, 6.2.16, Was Table 6-142–>Table 6-153
5.1	1126 Add _DSD Predefined Object– “DeviceSpecific Data” properties	Was Table 5-133 & 6-142 now–>5-148 & 6-157
5.1	1123 CPPC Performance Feedback Counter Change, 1130 CPPC2, [overlapping/duplicate tickets]	Tables 5-126, 8.4.5, 8.4.5.1x , 8.4.5.1, 8.4.5.1.3.1-4
5.1	1116 Add x2APIC and GIC structure for _MAT method	6.2.10
5.0 B	1145 Support GICs in proximity domain	5.2.16 5.2. new section 16.4 new tables, 6.2.13 Table 5-65
5.0 B	1144 Fix the gap for Notify value description	5.6.6, new tables: Table 5-132, 5-133
5.0 B	1142 Error Source Notifications	18.3.2.6.2, 18.4, Table 18-290
5.0 B	1117 Move http://acpi.info/links.htm content to UEFI Forum Website	1.10, 5.2.4, 5.2.22.3, 5.2.24, 5.6.7, 9.8.3.2, 13, 13.2.2 A.2.4, A.2.5, Tables 5-31, 5-60, 5-133

continues on next page

Table 2 – continued from previous page

5.0 B	1113 Typos in ACPI 5.0a	Table 6-184
5.0 B	1148 Inconsistent BIX object description/example	Was Table 10-234–>10-250
5.0 B	1143 Typos in ACPI 5.0a	6.1.8, 8.4.1
5.0 B	1102 Clarify Use of GPE Block Devices in Hardware-Reduced ACPI	3.11.1, 4.1, 9.10
5.0 B	Mantis 1114 Lack of description on Bit 4 of _STA	6.3.7
5.0 A	Jira 51 incorrect type information	Table 19-322
5.0 A	Jira 50 Misspelling of “management”	3.1
5.0 A	Jira 49 Updated description of DerefOf to specify behavior when attempt is made to de-reference a reference (via Index) to a NULL (empty) package element.	19.5.29
5.0 A	Jira 48 Text changes to change PM Timer from required to optional	4.8.1.4, 4.8.2.1, 4.8.3.3, 5.2.9
5.0 A	Jira 46 Figure 5-29 is a printer killer	Fig 5-29
5.0 A	Jira 45 Typos in Figure 5-30	Fig 5-30
5.0 A	Jira 44 Link issues in table 5-133	Table 5-133
5.0 A	Jira 43 Invalid AddressSpaced keywords in example ASL code, orphan _REG	6.5.4
5.0 A	Jira 42 Serious bug in ASL example code for _OSC	6.2.10.4
5.0 A	Jira 41 Fix problems with PCC address space description	14.5
5.0 A	Jira 40 Issues with _GRT and _SRT Buffer description	9.18.3, 9.18.4
5.0 A	Jira 39 Clarification needed for _CST	Table 8-206
5.0 A	Jira 38 Incorrect field name in “Generic Register Descriptor”.	6.4.3.7
5.0 A	Jira 37 Clarifications for _CPC method	8.4.5.1.2.1-2
5.0 A	Jira 36 Restore legality of module-level executable AML code.	19.1.3
5.0 A	Jira 35 ASL grammar: “UserTerm” is confusing	19.1
5.0 A	Jira 34 Description of _GTM has a bad line with very large font	9.8.2.1.1
5.0 A	Jira 33 Missing information in _CPC description	8.4.5.1
5.0 A	Jira3 2 Error in description of _REG method	6.5.4
5.0 A	Jira 31 Clarify length field for Serial resource descriptor	6.4.3.8.2, Table 6-190
5.0 A	Jira 30 Argument descriptions in incorrect order for resource descriptors	19.5.41, 19.5.101
5.0 A	Jira 29 Issues with memory descriptors (grammar and macros)	19.1, 19.5
5.0 A	Jira 28 Problems with ASL grammar entry for DWordMemory	19.1.8
5.0 A	Jira 27 Problems with Unicode description for _MLS method	6.1.7
5.0 A	Jira 26 Incorrect grammar for “32-bits” and “64-bits”	Throughout spec
5.0 A	Jira 25 Incorrect table reference in 19.2.5.4	19.2.5.4
5.0 A	Jira 24 Resource Descriptor tables – formatting issues	6.4
5.0 A	Jira 23 Interrupt Descriptors: Wake bit should be split from Share bit	6.4
5.0 A	Jira 22 ASL grammar for ObjectType operator is incorrect	19.1.6
5.0 A	Jira 21 ASL grammar is missing description of type 6 opcodes	19.1.5

continues on next page

Table 2 – continued from previous page

5.0 A	Jira 20 Problems with table 5-31 (reserved ACPI table signatures)	Table 5-31
5.0 A	Jira 19 Clarify description of _BQC method	B.5.4
5.0 A	Jira 18 Fix for EC OpRegion availability example	5.2.15
5.0 A	Jira 17 Clarify meaning of BGRT status field	Table 5-97
5.0 A	Jira 16 Correction to _DSM example	9.14.1
5.0 A	Jira 15 Clarify _DSM backward compatibility requirement and example	9.15.1
5.0 A	Jira 14 Description of _CPC is missing definition of unsupported optional registers	8.4.5.1
5.0 A	Jira 13 Incorrect _PLD name expansion	Table 5-133, 6.1.8
5.0 A	Jira 12 PLD description needs clarification	6.1.8
5.0 A	Jira 11 Errata forwarded from HP	5.2.24, 5.6.5.3
5.0 A	Jira 10 More issues with ACPI table 5-133	Table 5-133
5.0 A	Jira 7 Error in QWordIO, ExtendedIO descriptions	19.5.41, 19.5.101
5.0 A	Jira 6 Appendix A is now misnamed in ACPI 5.0	Appendix A
5.0 A	Jira 5 PARTIAL–Need group agreement–Method _GTS and _BFS are unused, should be removed from ACPI spec.	7.3, 7.3.3, 16.1, 16.1.6-7, fig. 7-204
5.0 A	Jira 4 Table 5-133 - issues with _Sx methods	Table 5-133
5.0 A	Jira 3 Issues with predefined names table (table 5-133)	Table 5-133
5.0 A	Jira 2 Description of new sleep control register incorrect	Table 4-24
5.0 A	Jira 1 SystemCMOS keyword inconsistencies	Table 5-114, 5.5.2.4.1, 6.5.4 19, 5.96, 9.15.1-2, 19.5.96, 20.2.5.2
5.0	Ptec-002	5.2.6
5.0	MSFT-020 Enumeration Power Controls	7.2.7, 7.2.12
5.0	MSFT-019 GTDT table	5.2.24
5.0	MSFT_0018 Locking Targets from AML	5.7.5
5.0	MSFT-0017 PLD clarification for handhelf form factors	5.1.8
5.0	MSFT-0016 Extended GPIO-signaled Event Numbers	5.6.5.3
5.0	MSFT-0015 (0.1) D3 Cold Errata	7.2.1, 7.2.18 through 7.2.22
5.0	MSFT-0014	5.2.23
5.0	MSFT-0013_ADR for SIO	6.2
5.0	MSFT-0012 ROM (Get ROM Data)	5.6.6, 9.16
5.0	MSFT-010 Reserved Table Signatures	5.2.6
5.0	MSFT-0009 (0.4)TimeAndAlarmDevice Modification	9.18
5.0	MSFT-0008 Collaborative Processor Performance Control	8.4.5
5.0	MSFT-0007 Platform Communications Channel added (new ch. 14)	Ch 14 (new)
5.0	MSFT-0007-0008 Platform Communication Channel and CPPC changes	Ch 14 (new)
5.0	MSFT-0006 SPB Abstraction	3.11.3, 5.5.2.4.5.x, 6.4.3.8.2, 6.5.8, 18.1.3, 18.1.6, 18.1.7, 18.5.44, 18.5x, 19.2.5.2
5.0	MSFT-0005 GPIO Abstraction	5.5.2.4.x, 5.6, 5.6.5.x, 6.4.3, 6.3.8.x, 18.5.51, 18.5.52, 18.5.89
5.0	MSFT-0004 (0.2) Fixed DMA Descriptor	6.4.2.9, 18.5.50
5.0	MSFT-0003 Device identification	6.1, 6.1.3, 6.1.5, 6.1.6, 6.1.9

continues on next page

Table 2 – continued from previous page

5.0	MSFT-0002 Interrupt Descriptors for Generic Interrupt Controller	5.2.11, 5.2.14-15
5.0	MSFT-0001 HW-reduced ACPI	3.11.x, 4, 4.1.x, 4.3.7, 5.2.9, 5.2.9.1, 6.4.2.1, 6.4.3.6, 7.2.11, 7.3.4, 9.6, 12, 12.1, 12.6, 12.11, 12.11.1, 15, 15.1.x, 15.3, 15.3.1.x, 18.5.55, 18.5.57
5.0	INTC-0014 Remove a line (reference) not needed	A.2.3
5.0	INTC-0013	n/a
5.0	INTC-0012 fix AML opcode table	19.3
5.0	INTC-0011 fix table offsets	18.6.x (tables)
5.0	INTC-0010 Update Constant Descriptions	18.5.88, 18.5.89, 18.5.104, 18.5.136
5.0	INTC0009 RASF	5.2.20.x
5.0	INTC-008	5.2.6
5.0	INTC-006 Fixed Example	6.2.10.4
5.0	INTC-005 Update Package Description	18.5.92
5.0	INTC-004 Table Definition Language	20, 21.x
5.0	INTC-003 MPST	6.1, 6.1.3, 6.1.5, 6.1.6, 6.1.9
5.0	INTC-002 EINJ	17.6.1, 17.6.3, 17.6.5
5.0	INTC-001 (0.8) Firmware Performance Data Table (FPDT)	5.2.20.4, 5.2.20.6
5.0	INTC-001 Firmware Performance Data Table (FPDT) (0.4)	5.2.19- 5.2.20.6
5.0	HP-0002 Additional Hardware Error Notification Types	18.3.2.7
5.0	HP-0001 (0.2) BMC Requested Graceful Shutdown	5.6.5, 6.3.5
5.0	ACPI4.0 _DSM function 0 clarification	9.14.1
5.0	AMD-002 0.3 ROM (Get ROM Data)	B.3.3
4.0a	Errata corrected and clarifications added.	2.2, 5.2.6, 5.2.12.4, 5.2.18, 5.5.2.4.3.1, 5.6.5, 5.6.6, 5.6.7, 6.4.2.8, 6.4.3.5.1-3, 6.5.7, 8.4.3.4, 8.4.4.5, 8.4.5, 9.2.5, 9.8.2.1.1, 9.10, 9.13, 10.4.1, 10.1.3.1, 10.2.2, 10.2.1.1-2, 10.2.2.8, 10.2.2.9, , 10.3, 10.3.3, 10.4, 10.3.4, 10.4.1, 10.5, 15.1, 17.1, 17.3.1, 17.3.2.6.1, 17.3.2.6.2, 17.4, 17.5.1.1, 17.6.1, 17.6.3, 18.1.8, 18.5.44, 18.5.89, 18.5.101
4.0	Major specification revision. Clock Domains, x2APIC Support, Logical Processor Idling, Corrected Platform Error Polling Table, Maximum System Characteristics Table, Power Metering and Budgeting, IPMI Operation Region, USB3 Support in _PLD, Re-evaluation of _PPC acknowledgement via _OST, Thermal Model Enhancements, _OSC at _SB, Wake Alarm Device, Battery Related Extensions, Memory Bandwidth Monitoring and Reporting, ACPI Hardware Error Interfaces, D3hot.	n/a
3.0b	Errata corrected and clarifications added.	n/a
3.0a	Errata corrected and clarifications added.	n/a

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Table 2 – continued from previous page

3.0	Major specification revision. General configuration enhancements. Inter-Processor power, performance, and throttling state dependency support added. Support for > 256 processors added. NUMA Distancing support added. PCI Express support added. SATA support added. Ambient Light Sensor and User Presence device support added. Thermal model extended beyond processor-centric support.	n/a
2.0c	Errata corrected and clarifications added.	n/a
2.0b	Errata corrected and clarifications added.	n/a
2.0a	Errata corrected and clarifications added. ACPI 2.0 Errata Document Revision 1.0 through 1.5 integrated.	n/a
2.0 Errata Rev. 1.5	Errata corrected and clarifications added.	n/a
2.0 Errata Rev. 1.4	Errata corrected and clarifications added.	n/a
2.0 Errata Rev. 1.3	Errata corrected and clarifications added.	n/a
2.0 Errata Rev. 1.2	Errata corrected and clarifications added.	n/a
2.0 Errata Rev. 1.1	Errata corrected and clarifications added.	n/a
2.0 Errata Rev. 1.0	Errata corrected and clarifications added.	n/a
2.0	Major specification revision. 64-bit addressing support added. Processor and device performance state support added. Numerous multiprocessor workstation and server-related enhancements. Consistency and readability enhancements throughout.	n/a
1.0b	Errata corrected and clarifications added. New interfaces added.	n/a
1.0a	Errata corrected and clarifications added. New interfaces added.	n/a
1.0	Original Release.	n/a

Overview

The following provides a high-level overview of the Advanced Configuration and Power Interface (ACPI). To make it easier to understand ACPI, this section focuses on broad and general statements about ACPI and does not discuss every possible exception or detail about ACPI. The rest of the ACPI specification provides much greater detail about the inner workings of ACPI than is discussed here, and is recommended reading for developers using ACPI.

History of ACPI

ACPI was developed through collaboration between Intel, Microsoft*, Toshiba*, HP*, and Phoenix* in the mid-1990s. Before the development of ACPI, operating systems (OS) primarily used BIOS (Basic Input/Output System) interfaces for power management and device discovery and configuration. This power management approach used the OS's ability to call the system BIOS natively for power management. The BIOS was also used to discover system devices and load drivers based on probing input/output (I/O) and attempting to match the correct driver to the correct device (plug and play). The location of devices could also be hard coded within the BIOS because the platform itself was non-enumerable. These solutions were problematic in three key ways. First, the behavior of OS applications could be negatively affected by the BIOS-configured power management settings, causing systems to go to sleep during presentations or other inconvenient times. Second, the power management interface was proprietary on each system. This required developers to learn how to configure power management for each individual system. Finally, the default settings for various devices could also conflict with each other, causing devices to crash, behave erratically, or become undiscoverable.

ACPI was developed to solve these problems and others.

What is ACPI?

ACPI can first be understood as an architecture-independent power management and configuration framework that forms a subsystem within the host OS. This framework establishes a hardware register set to define power states (sleep, hibernate, wake, etc). The hardware register set can accommodate operations on dedicated hardware and general purpose hardware.

The primary intention of the standard ACPI framework and the hardware register set is to enable power management and system configuration without directly calling firmware natively from the OS. ACPI serves as an interface layer between the operating system and system firmware, as shown in figure i-1.

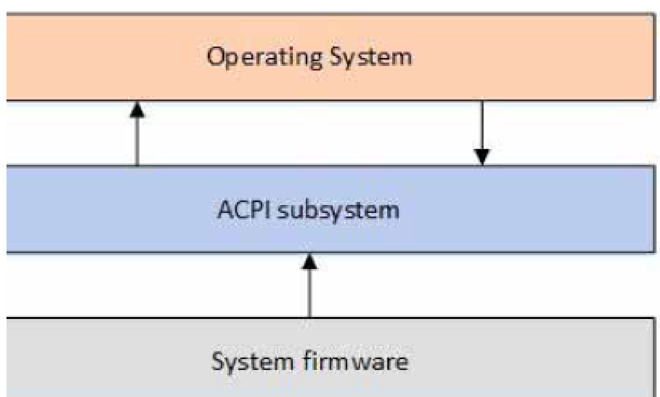

Fig. 1: *Fig. i-1 - ACPI overview*

ACPI defines two types of data structures that are shared between system firmware and the OS via the ACPI subsystem: data tables and definition blocks (see figure i-2). These data structures are the primary communication mechanism between the firmware and the OS. Data tables store raw data and are consumed by device drivers. Definition blocks consist of byte code that is executable by an interpreter.

Upon initialization, the AML interpreter extracts byte code in the definition blocks as enumerable objects. This collection of enumerable objects forms an OS construct called the ACPI namespace. Objects in the ACPI namespace can

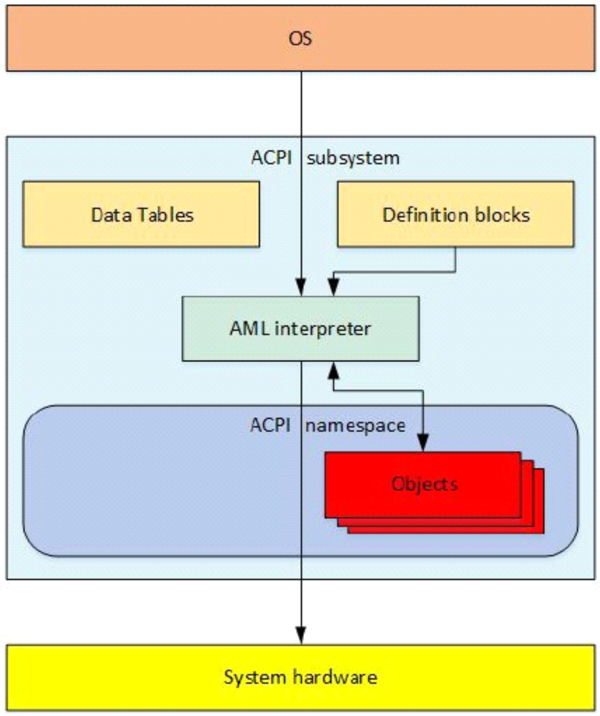

Fig. 2: *Fig. i-2 - ACPI Structure*

either have a directly defined value, or be evaluated by the AML interpreter. The AML interpreter, directed by the OS, evaluates objects and then interfaces with system hardware to perform necessary operations.

The definition block byte code is compiled from the ACPI Source Language (ASL) code. ASL is the language used to define ACPI objects and to write control methods. An ASL compiler translates ASL into ACPI Machine Language (AML) byte code. AML is the language processed by the AML interpreter, as shown in figure i-3.

ACPI Source Language (ASL) code is used to define objects and control methods. Then the ASL compiler translates ASL into ACPI Machine Language (AML) byte code contained within ACPI definition Blocks. Definition blocks consist of an identifying table header and byte code that is executable by an AML interpreter.

The AML interpreter executes byte code and evaluates objects in the definition blocks to allow the byte code to perform loop constructs, conditional evaluations, access defined address spaces, and perform other operations that applications require. The AML interpreter has read/write access to defined address spaces, including system memory, I/O, PCI configuration, and more. It accesses these address spaces by defining entry points called objects. Objects can either have a directly defined value or else must be evaluated and interpreted by the AML interpreter.

This collection of enumerable objects is an OS construct called the ACPI namespace. The namespace is a hierarchical representation of the ACPI devices on a system. The system bus is the root of enumeration for these ACPI devices. Devices that are enumerable on other buses, like PCI or USB devices, are usually not enumerated in the namespace. Instead, their own buses enumerate the devices and load their drivers. However, all enumerable buses have an encoding technique that allows ACPI to encode the bus-specific addresses of the devices so they can be found in ACPI, even though ACPI usually does not load drivers for these devices.

Generally, devices that have a _HID object (hardware identification object) are enumerated and have their drivers loaded by ACPI. Devices that have an _ADR object (physical address object) are usually not enumerated by ACPI and generally do not have their drivers loaded by ACPI. _ADR devices usually can perform all necessary functions without involving ACPI, but in cases where the device driver cannot perform a function, or if the driver needs to communicate to system firmware, ACPI can evaluate objects to perform the needed function.

As an example of this, PCI does not support native hotplug. However, PCI can use ACPI to evaluate objects and define

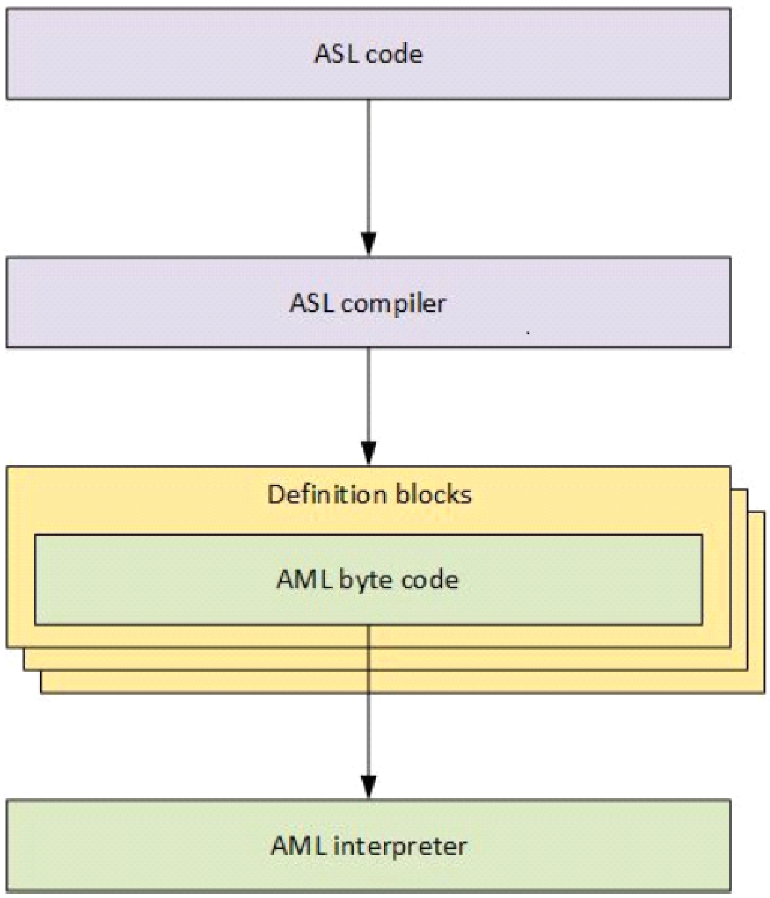

Fig. 3: *Fig. i-3 - ASL and AML*

methods that allow ACPI to fill in the functions necessary to perform hotplug on PCI.

An additional aspect of ACPI is a runtime model that handles any ACPI interrupt events that occur during system operation. ACPI continues to evaluate objects as necessary to handle these events. This interrupt-based runtime model is discussed in greater detail in the Runtime model section below.

ACPI Initialization

The best way to understand how ACPI works is chronologically. The moment the user powers up the system, the system firmware completes its setup, initialization, and self tests.

The system firmware then uses information obtained during firmware initialization to update the ACPI tables as necessary with various platform configurations and power interface data, before passing control to the bootstrap loader. The extended root system description table (XSDT) is the first table used by the ACPI subsystem and contains the addresses of most of the other ACPI tables on the system. The XSDT points to the fixed ACPI description table (FADT) as well as other major tables that the OS processes during initialization. After the OS initializes, the FADT directs the ACPI subsystem to the differentiated system description table (DSDT), which is the beginning of the namespace because it is the first table that contains a definition block.

The ACPI subsystem then processes the DSDT and begins building the namespace from the ACPI definition blocks. The XSDT also points to the secondary system description tables (SSDTs) and adds them to the namespace. The ACPI data tables give the OS raw data about the system hardware.

After the OS has built the namespace from the ACPI tables, it begins traversing the namespace and loading device drivers for all the _HID devices it encounters in the namespace. See figure i-4.

In the ACPI Initialization diagram above, system firmware updates the ACPI tables as necessary with information only available at runtime, before handing off control to the bootstrap loader. The XSDT is the first table used by the OS's ACPI subsystem, and contains addresses of most other ACPI tables on the system. The XSDT points to the FADT, the SSDTs, and other major ACPI tables. The FADT directs the ACPI subsystem to the DSDT, which is the beginning of the namespace because DSDT is the first table that contains a definition block. The ACPI subsystem then consumes the DSDT and begins building the ACPI namespace from the definition blocks. The XSDT also points to the SSDTs and adds them to the namespace.

Runtime Model

After the system is up and running, ACPI works with the OS to handle any ACPI events that occur via an interrupt. This interrupt invokes ACPI events in one of two general ways: fixed events and general purpose events (GPEs).

Fixed events are ACPI events that have a predefined meaning in the ACPI specification. These fixed events include actions like pressing the power button or ACPI timer overflows. These events are handled directly by the OS handlers.

GPEs are ACPI events that are not predefined by the ACPI specification. These events are usually handled by evaluating control methods, which are objects in the namespace and can access system hardware. When the ACPI subsystem evaluates the control method with the AML interpreter, the GPE object handles the events according to the OS's implementation. Typically this might involve issuing a notification to a device to invoke the device driver to perform a function.

We discuss a generic example of this runtime model in the next section.

Thermal Event Example

ACPI includes a thermal model to allow systems to control the system temperature either actively (by performing actions like turning a fan on) or passively by reducing the amount of power the system uses (by performing actions like throttling the processor). We can use an example of a generic thermal event shown in Figure i-5 to demonstrate how the ACPI runtime model works.

The ACPI thermal zone includes control methods to read the current system temperature and trip points.

When the OS initially finds a thermal zone in the namespace, it loads the thermal zone driver, which evaluates the thermal zone to obtain the current temperature and trip points.

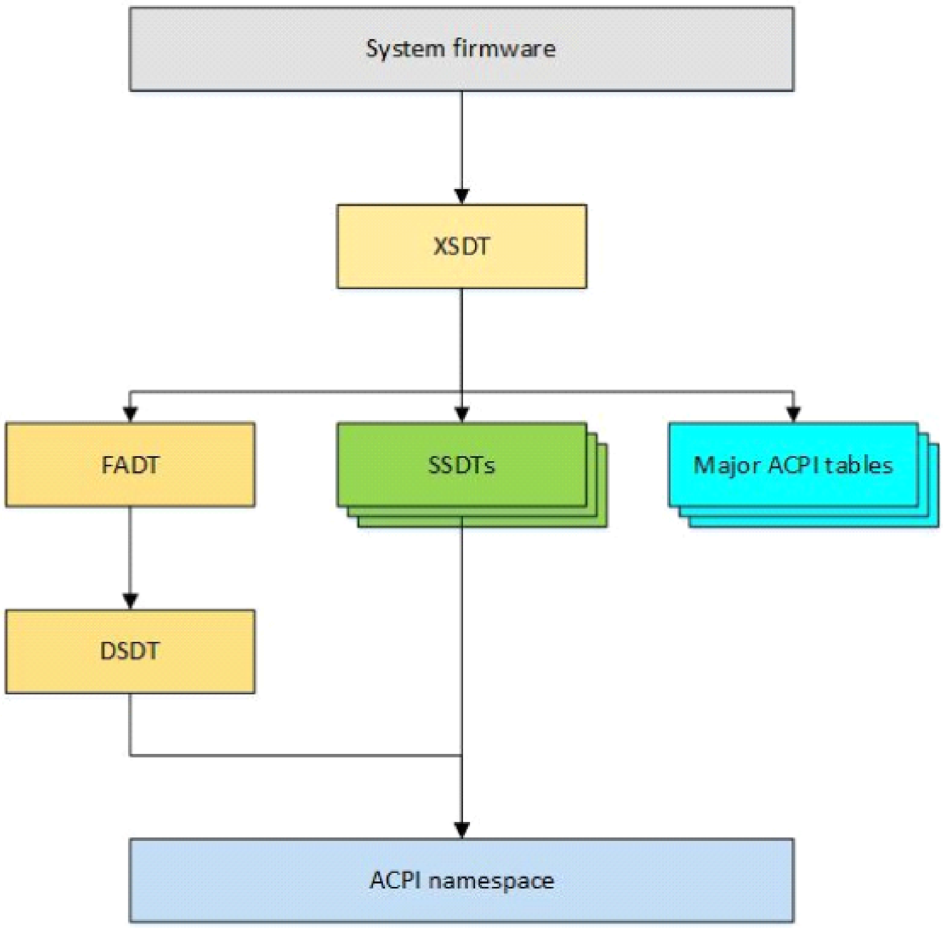

Fig. 4: *Fig. i-4 ACPI Initialization*

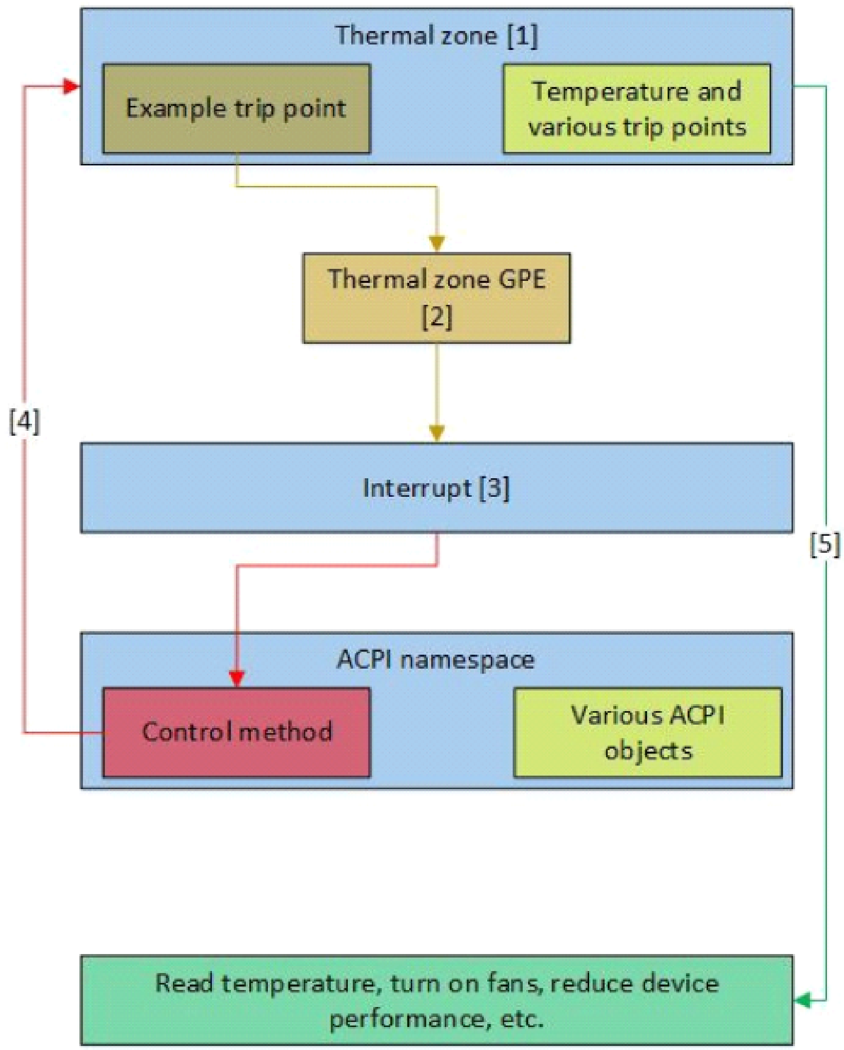

Fig. 5: **Fig. i-5 Runtime Thermal Event**

When a system component heats up enough to trigger a trip point, a thermal zone GPE occurs.

The GPE causes an interrupt to occur. When the ACPI subsystem receives the interrupt, it first checks whether any fixed events have occurred. In this example, the thermal zone event is a GPE, so no fixed event has occurred.

The ACPI subsystem then searches the namespace for the control method that matches the GPE number of the interrupt. Upon finding it, the ACPI subsystem evaluates the control method, which might then access hardware and/or notify the thermal zone handler.

The operating system's thermal zone handler then takes whatever actions are necessary to handle the event, including possibly accessing hardware.

ACPI is a very robust interface implementation. The thermal zone trip point could notify the system to turn on a fan, reduce a device's performance, read the temperature, shut down the system, or any combination of these and other actions depending on the need.

This runtime model is used throughout the system to manage all of the ACPI events that occur during system operation.

Summary

ACPI can best be described as a framework of concepts and interfaces that are implemented to form a subsystem within the host OS. The ACPI tables, handlers, interpreter, namespace, events, and interrupt model together form this implementation of ACPI, creating the ACPI subsystem within the host OS. In this sense, ACPI is the interface between the system hardware/firmware and the OS and OS applications for configuration and power management. This gives various OS a standardized way to support power management and configuration via the ACPI namespace.

The ACPI namespace is the enumerable, hierarchical representation of all ACPI devices on the system and is used to both find and load drivers for ACPI devices on the system. The namespace can be dynamic by evaluating objects and sending interrupts in real time, all without the need for the OS to call native system firmware code. This enables device manufacturers to code their own instructions and events into devices. It also reduces incompatibility and instability by implementing a standardized power management interface.